

Mathematics I

BCA — Semester 1 — 2018 — Answer Sheet

Subject Code: CACS104

Group B

Attempt any SIX questions. $[6 \times 5 = 30]$

11. 32 students play basketball and 25 students play volleyball. It is found that 20 students play both the games. Find the number of students playing at least one game. Also, find total number of students if 13 students play none of these games.

[5]

Answer: Solution:

Let B = set of basketball players, V = set of volleyball players.

Given: $n(B) = 32$, $n(V) = 25$, $n(B \cap V) = 20$

Students playing at least one game:

$$n(B \cup V) = n(B) + n(V) - n(B \cap V)$$

$$n(B \cup V) = 32 + 25 - 20 = \mathbf{37 \text{ students}}$$

Total number of students:

$$\text{Total} = n(B \cup V) + \text{students playing none}$$

$$\text{Total} = 37 + 13 = \mathbf{50 \text{ students}}$$

12. Let $f : \mathbb{N} \rightarrow \mathbb{N}$ be defined by $f(x) = 2x$ for all $x \in \mathbb{N}$ where $\mathbb{N}$ is the set of natural numbers. Show that f is one-one but not onto function.

[5]

Answer: Solution:

Given: $f(x) = 2x$, where $f : \mathbb{N} \rightarrow \mathbb{N}$

To prove f is one-one (injective):

Let $f(x_1) = f(x_2)$

Then $2x_1 = 2x_2$

Therefore $x_1 = x_2$

Hence, f is one-one.

To prove f is not onto (surjective):

For f to be onto, every element in the codomain $\mathbb{N}$ must have a pre-image in the domain.

Consider $y = 3 \in \mathbb{N}$ (codomain).

If $f(x) = 3$, then $2x = 3$, so $x = 1.5 \notin \mathbb{N}$.

Since 3 has no pre-image in $\mathbb{N}$, f is **not onto**.

Therefore, f is one-one but not onto.

13. If the three consecutive terms of a geometric series be increased by their middle term, then prove that the resulting terms will be in harmonic progression (H.P.).

[5]

Answer: Proof:

Let the three consecutive terms of a G.P. be a/r , a , and ar .

When each term is increased by the middle term a , the new terms are:

- First term: $a/r + a = a(1 + r)/r$
- Second term: $a + a = 2a$
- Third term: $ar + a = a(r + 1)$

For these to be in H.P., their reciprocals must be in A.P.

Reciprocals:

- $1/[a(1+r)/r] = r/[a(1+r)]$
- $1/(2a)$
- $1/[a(1+r)]$

Check if they are in A.P.:

$$2 \times [1/(2a)] = 1/a$$

$$r/[a(1+r)] + 1/[a(1+r)] = (r+1)/[a(1+r)] = 1/a$$

Since $2 \times (\text{second reciprocal}) = \text{first} + \text{third}$, the reciprocals are in A.P.

Therefore, the resulting terms are in **Harmonic Progression**.

14. Find the adjoint of the matrix:

$$A = \begin{bmatrix} 1 & 2 & -2 \\ 2 & 2 & 1 \\ 0 & 3 & -1 \end{bmatrix}$$

[5]

Answer: Solution:

Given matrix $A = \begin{bmatrix} 1 & 2 & -2 \\ 2 & 2 & 1 \\ 0 & 3 & -1 \end{bmatrix}$

Step 1: Find the cofactor matrix.

$$C_{11} = +(2 \times (-1) - 1 \times 3) = -5$$

$$C_{12} = -(2 \times (-1) - 1 \times 0) = 2$$

$$C_{13} = +(2 \times 3 - 2 \times 0) = 6$$

$$C_{21} = -(2 \times (-1) - (-2) \times 3) = -4$$

$$C_{22} = +(1 \times (-1) - (-2) \times 0) = -1$$

$$C_{23} = -(1 \times 3 - 2 \times 0) = -3$$

$$C_{31} = +(2 \times 1 - (-2) \times 2) = 6$$

$$C_{32} = -(1 \times 1 - (-2) \times 2) = -5$$

$$C_{33} = +(1 \times 2 - 2 \times 2) = -2$$

Cofactor matrix = $\begin{bmatrix} -5 & 2 & 6 \\ -4 & -1 & -3 \\ 6 & -5 & -2 \end{bmatrix}$

Step 2: Take the transpose to get the adjoint.

$$\text{adj}(A) = \begin{bmatrix} -5 & -4 & 6 \\ 2 & -1 & -5 \\ 6 & -3 & -2 \end{bmatrix}$$

15. Prove that:

$$\begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ x^2 & y^2 & z^2 \end{vmatrix} = (x-y)(y-z)(z-x)$$

[5]

Answer: Proof:

$$\text{Let } \Delta = \begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ x^2 & y^2 & z^2 \end{vmatrix}$$

Apply $C_1 \rightarrow C_1 - C_2$ and $C_2 \rightarrow C_2 - C_3$:

$$\Delta = \begin{vmatrix} 1 & 0 & 0 \\ x & y-x & z-x \\ x^2 & y^2-x^2 & z^2-x^2 \end{vmatrix}$$

Expand along R_1 :

$$\Delta = 1 \times [(y-x)(z^2-x^2) - (z-x)(y^2-x^2)]$$

$$\Delta = (y-x)(z-x)(z+x) - (z-x)(y-x)(y+x)$$

$$\Delta = (y-x)(z-x)[(z+x) - (y+x)]$$

$$\Delta = (y-x)(z-x)(z-y)$$

$$\Delta = (x-y)(y-z)(z-x)$$

Hence proved.

16. Find the equation of parabola with focus $(-1, 2)$ and directrix $x = 5$.

[5]

Answer: Solution:

Given: Focus $S = (-1, 2)$, Directrix: $x = 5$

Let $P(x, y)$ be any point on the parabola.

By definition, $SP = PM$ (distance from focus = perpendicular distance from directrix)

$$SP^2 = (x + 1)^2 + (y - 2)^2$$

$$PM = |x - 5|$$

$$PM^2 = (x - 5)^2$$

Since $SP^2 = PM^2$:

$$(x + 1)^2 + (y - 2)^2 = (x - 5)^2$$

$$x^2 + 2x + 1 + y^2 - 4y + 4 = x^2 - 10x + 25$$

$$2x + 1 + y^2 - 4y + 4 = -10x + 25$$

$$12x + y^2 - 4y - 20 = 0$$

Equation of parabola: $y^2 - 4y + 12x - 20 = 0$

$$\text{or } (y - 2)^2 = -12(x - 2)$$

17. Transform $u = [[-1, 3], [2, 1]]$ and $v = [[0, 1], [-1, 0]]$ and check whether this transformation is linear.

[5]

Answer: Solution:

Given: $u = [[-1, 3], [2, 1]]$ and $v = [[0, 1], [-1, 0]]$

Wait, the question seems unclear. If the transformation is defined by $T(x) = ux + v$, then:

Check for linearity:

A transformation T is linear if:

1. $T(x + y) = T(x) + T(y)$

2. $T(cx) = cT(x)$

If $T(x) = ux + v$ where $v \neq 0$, then:

$$T(0) = u(0) + v = v \neq 0$$

Since $T(0) \neq 0$, the transformation is **not linear**.

For a transformation to be linear, it must satisfy $T(0) = 0$. The presence of the translation vector v makes this an affine transformation, not a linear transformation.

Group C

Attempt any TWO questions. $[2 \times 10 = 20]$

18. Define permutation and combination. Try to establish relationship between them with the help of formulae. In how many ways can the letters of the word 'LOGIC' be arranged so that i) Vowels may occupy odd position? ii) No vowels are together?

[10]

Answer: Permutation: An arrangement of objects in a specific order. The number of permutations of n objects taken r at a time is:

$$P(n, r) = \frac{n!}{(n-r)!}$$

Combination: A selection of objects without regard to order. The number of combinations of n objects taken r at a time is:

$$C(n, r) = \frac{n!}{r!(n-r)!}$$

Relationship: $P(n, r) = C(n, r) \times r!$

Word 'LOGIC':

Letters: L, O, G, I, C (5 letters, vowels: O, I; consonants: L, G, C)

i) Vowels in odd positions:

Odd positions: 1, 3, 5 (3 positions)

Even positions: 2, 4 (2 positions)

Arrange 2 vowels in 3 odd positions: $P(3, 2) = \frac{3!}{(3-2)!} = 6$ ways

Arrange 3 consonants in 3 remaining positions: $3! = 6$ ways

Total ways = $6 \times 6 = 36$ ways

ii) No vowels together:

First arrange 3 consonants: $3! = 6$ ways

This creates 4 gaps: _C_C_C_

Choose 2 gaps from 4 for vowels: $C(4, 2) = 6$

Arrange 2 vowels in these 2 gaps: $2! = 2$

Total ways = $6 \times 6 \times 2 = 72$ ways

19. Define scalar and vector product in three dimensional space with their geometrical interpretation and prove the formula $\sin(A + B) = \sin A \cos B + \cos A \sin B$ by using vector method.

[10]

Answer: Scalar Product (Dot Product):

For vectors $\mathbf{a}$ and $\mathbf{b}$, $\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}|\cos \theta$

Geometrical interpretation: The scalar product gives the product of the magnitude of one vector and the projection of the other vector onto it.

Vector Product (Cross Product):

For vectors $\mathbf{a}$ and $\mathbf{b}$, $\mathbf{a} \times \mathbf{b} = |\mathbf{a}||\mathbf{b}|\sin \theta \mathbf{n}$

where $\mathbf{n}$ is the unit vector perpendicular to both $\mathbf{a}$ and $\mathbf{b}$.

Geometrical interpretation: The magnitude of the cross product equals the area of the parallelogram formed by the two vectors.

Proof of $\sin(A+B) = \sin A \cos B + \cos A \sin B$:

Consider unit vectors in the XY-plane:

$\mathbf{a} = \cos A \mathbf{i} + \sin A \mathbf{j}$ (making angle A with X-axis)

$\mathbf{b} = \cos B \mathbf{i} - \sin B \mathbf{j}$ (making angle $-B$ with X-axis)

The angle between $\mathbf{a}$ and $\mathbf{b}$ is $(A + B)$.

$$|\mathbf{a} \times \mathbf{b}| = |\mathbf{a}||\mathbf{b}|\sin(A+B) = \sin(A+B) \dots (i)$$

$$\text{Also, } \mathbf{a} \times \mathbf{b} = (\cos A \mathbf{i} + \sin A \mathbf{j}) \times (\cos B \mathbf{i} - \sin B \mathbf{j})$$

$$= -\cos A \sin B (\mathbf{i} \times \mathbf{j}) + \sin A \cos B (\mathbf{j} \times \mathbf{i})$$

$$= -\cos A \sin B (\mathbf{k}) + \sin A \cos B (-\mathbf{k})$$

Wait, let me recalculate:

$$\mathbf{a} \times \mathbf{b} = \cos A (-\sin B) (\mathbf{i} \times \mathbf{j}) + \sin A (\cos B) (\mathbf{j} \times \mathbf{i})$$

$$= -\cos A \sin B (\mathbf{k}) + \sin A \cos B (-\mathbf{k})$$

$$= -(\cos A \sin B + \sin A \cos B) \mathbf{k}$$

$$\text{Taking magnitude: } |\mathbf{a} \times \mathbf{b}| = \cos A \sin B + \sin A \cos B \dots (ii)$$

From (i) and (ii):

$$\sin(A+B) = \sin A \cos B + \cos A \sin B$$

Hence proved.

20. Define the logarithmic function, state its properties and if $f(x) = \log\left[\frac{1+x}{1-x}\right]$ for $-1 < x < 1$, show that $f(a) + f(b) = f\left[\frac{a+b}{1+ab}\right]$.

[10]

Answer: Logarithmic Function:

The logarithmic function $f(x) = \log_a(x)$ is the inverse of the exponential function. It is defined for $x > 0$ and $a > 0, a \neq 1$.

Properties:

1. $\log_a(mn) = \log_a m + \log_a n$
2. $\log_a(m/n) = \log_a m - \log_a n$
3. $\log_a(m^n) = n \cdot \log_a m$
4. $\log_a 1 = 0$
5. $\log_a a = 1$
6. $a^{(\log_a x)} = x$

Proof:

Given: $f(x) = \log\left[\frac{1+x}{1-x}\right]$

$$\begin{aligned} \text{LHS} &= f(a) + f(b) \\ &= \log\left[\frac{1+a}{1-a}\right] + \log\left[\frac{1+b}{1-b}\right] \\ &= \log\left[\frac{(1+a)(1+b)}{(1-a)(1-b)}\right] \\ &= \log\left[\frac{1+a+b+ab}{1-a-b+ab}\right] \end{aligned}$$

$$\begin{aligned} \text{RHS} &= f\left[\frac{a+b}{1+ab}\right] \\ &= \log\left[\frac{1 + \frac{a+b}{1+ab}}{1 - \frac{a+b}{1+ab}}\right] \\ &= \log\left[\frac{(1+ab+a+b)/(1+ab)}{(1+ab-a-b)/(1+ab)}\right] \\ &= \log\left[\frac{1+a+b+ab}{1-a-b+ab}\right] \end{aligned}$$

Since $\text{LHS} = \text{RHS}$,

$$f(a) + f(b) = f\left[\frac{a+b}{1+ab}\right]$$

Hence proved.

Note: These answers are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.